MILITARY SPECIFICATION

THEOPHYLLINE, EPHEDRINE SULFATE, AND HYDROXYZINE HYDROCHLORIDE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Theophylline, Ephedrine Sulfate, and Hydroxyzine Hydrochloride Tablets.

1.2 Classification. The tablets covered by this specification shall be the following sizes (See 6.2):

Size 1	Bottle of 100s
Size 2	Bottle of 500s

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

Federal

FED-STD-140	Tablets (for Medicinal Purposes).

FSC 6505

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Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Theophylline, Ephedrine Sulfate, and Hydroxyzine Hydrochloride Tablets, and shall be in accordance with all applicable requirements of FED-STD-140 and as specified herein:

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S2. Classification. Shall be type I.

S5.3 The following additional requirements and tests are added to this paragraph:

Shall contain the following formula per tablet, within the designated assay limits:

Theophylline (anhydrous) - - - - - - - - - -	130 mg
Ephedrine Sulfate- - - - - - - - - - - - - -	25 mg
Hydroxyzine Hydrochloride- - - - - - - - - -	10 mg

Tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of Ephedrine Sulfate and Hydroxyzine Hydrochloride, and not less than 90.0 percent and not more than 120.0 percent of the required amount of anhydrous Theophylline, when assayed as specified in 4.3.1 for Ephedrine Sulfate, 4.3.2 for Hydroxyzine Hydrochloride and 4.3.3 for anhydrous Theophylline.

The Anhydrous Theophylline and Ephedrine Sulfate used in the manufacture of the tablets shall be in accordance with the tests, standards, and requirements of the U.S.P. The Hydroxyzine Hydrochloride used in the manufacture of the tablets shall be in accordance with the tests, standards, and requirements of the N.F.

All other ingredients used in the manufacture of the tablets shall comply with S5.1.

S6.4.9.1 Disintegration. Tablets shall disintegrate in not more than 60 minutes.

3.2 Expiration dating period. The expiration dating period shall be 60 months.

4. REQUIREMENTS

4.1 Supplier Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period equal to the expiration dating period (potency period) specified by the contractor for the item(s) delivered under the contract but in no event less than 5 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-140.

4.3 Tests.

4.3.1 Ephedrine Sulfate assay.

Reagents.

Carbon disulfide, A.R.
Pyridine, spectrographic grade
Isopropyl alcohol, A.R.
Benzene, A.R.
Methyl alcohol anhydrous, A.R.
Glacial acetic acid, A.R.
Cupric chloride dihydrate ($CuCl_2.2H_2O$), A.R.
Sodium bicarbonate, A.R.
Methanol solution, aqueous, 10 percent - Mix 100 ml of methanol with 900 ml of purified water. Prepare fresh daily.
Acetic acid solution, aqueous, 10 percent - Mix 20 ml of glacial acetic acid with 180 ml of purified water.
Sodium bicarbonate solution, aqueous, 0.4 percent (w/v) - Weigh 4.0 grams of sodium bicarbonate into a 1000-ml volumetric flask, dissolve in and dilute to volume with purified water. Prepare fresh daily.
Carbon disulfide - pyridine reagent - Accurately measure and mix 25 ml of pyridine and 65 ml of isopropyl alcohol. Immediately prior to use add 35 ml of carbon disulfide and mix well.
Cupric chloride solution. Dissolve 60 mg of cupric chloride in 125 ml of purified water and dilute to 250 ml with pyridine.

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Procedure.

Accurately weigh approximately 70 mg of N.F. Ephedrine Sulfate Reference Standard into a 100-ml volumetric flask, dissolve in and dilute to volume with purified water. Pipet 10.0 ml into a 25-ml volumetric flask and dilute to volume with 0.4 percent sodium bicarbonate solution. Prepare fresh daily.

Accurately weigh the equivalent of approximately 70 mg of ephedrine sulfate from the sample preparation powder into a 200-ml volumetric flask. Add exactly 100.0 ml of 10 percent methanol solution and agitate the mixture on a mechanical shaker for 10 minutes. Allow to settle for approximately 10 minutes. Filter the supernatant solution through glass fiber filter paper (Whatman GF/C or equivalent), discarding the first 15 ml of filtrate. Pipet a 10.0 ml aliquot into a 25-ml volumetric flask and dilute to volume with 0.4 percent sodium bicarbonate solution.

To three separate 40-50 ml glass stoppered centrifuge tubes, add 3.0 ml aliquots of the final standard and sample solutions and a blank consisting of 3.0 ml of the 0.4 percent sodium bicarbonate solution. To each tube add 10 ml of carbon disulfide-pyridine reagent. Start timing the reaction. Then in succession, add 5 ml of the cupric chloride reagent and 6 ml of the 10 percent acetic acid solution, taking care to stopper and agitate the solution by inverting it four times. Centrifuge the tubes until the upper phase is clear. Carefully pipet 10.0 ml of the upper phase into a 25-ml volumetric flask and dilute to volume with isopropyl alcohol. After a lapse of exactly 35 minutes, measure the absorbances of the standard and sample solutions in a 1-cm cell with a suitable spectrophotometer at 440 mu, using the blank solution to set the instrument.

Calculation:

$$\frac{Au}{As} \times \frac{\text{mg Standard}}{\text{mg Sample}} \times \frac{\text{Average tablet weight (mg)}}{1} = \text{mg Ephedrine Sulfate per tablet.}$$

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$$\frac{\text{mg Ephedrine Sulfate} \times 100}{25} = \text{Percent required amount}$$

Where: Au = absorbance of sample
As = absorbance of standard

4.3.2 Hydroxyzine Hydrochloride assay.

Reagents:

Sodium hydroxide solution, 1N
Chloroform, A.R.
Hydrochloric acid solution, 0.1N
Anhydrous sodium sulfate, A.R.

Procedure.

Accurately weigh approximately 80 mg of N.F. Hydroxyzine Hydrochloride Reference Standard, previously dried at 105° C. for 2 hours, into a 200-ml volumetric flask. Dissolve in and dilute to volume with 0.1N hydrochloric acid solution. Pipet a 4.0 ml aliquot into a 100-ml volumetric flask and dilute to volume with 0.1N hydrochloric acid solution.

Accurately weigh the equivalent of approximately 16 mg of hydroxyzine hydrochloride from the sample preparation powder into a 100-ml volumetric flask containing approximately 50 ml of chloroform and several glass beads. After shaking mechanically for 45 minutes (wash down neck of flask with small portion of chloroform during shaking process), dilute to volume with chloroform. Pipet a 20.0 ml aliquot into a 125-ml separatory funnel containing 20 ml of 1N NaOH. Shake for two minutes and filter the chloroform phase through a shallow layer of anhydrous sodium sulfate into a 50-ml volumetric flask. Extract the sodium hydroxide layer with 20 ml of chloroform for 1 minute and filter the chloroform into the 50-ml flask. Rinse the layer of sodium sulfate with chloroform and dilute to volume with chloroform.

Pipet 10.0 ml of the chloroform solution into a 50-ml glass-stoppered Erlenmeyer flask and carefully evaporate to dryness on the steam bath with the aid of a stream of air or nitrogen. Cool and dissolve the residue in exactly 40.0 ml of 0.1N hydrochloric acid.

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Determine the absorbances of the final standard and sample solutions in 1-cm cells with a suitable spectrophotometer at 232 mu and 250 mu, using 0.1N hydrochloric acid as the blank solution.

Calculation:

$$\frac{Au}{As}\left(\frac{\text{mg Standard}}{200}\times\frac{4}{100}\right)\left(\frac{100}{\text{mg Sample}}\times\frac{50}{20}\times\frac{40}{10}\right)\frac{\text{Average wt/tablet(mg)}}{1}$$

= mg Hydroxyzine Hydrochloride per tablet

$$\frac{\text{mg Hydroxyzine Hydrochloride} \times 100}{10} = \text{percent required amount}$$

Where: Au = Absorbance sample at 232 mu minus absorbance at 250 mu.
As = Absorbance standard at 232 mu minus absorbance at 250 mu.

4.3.3 Theophylline assay.

Accurately weigh approximately 120 mg of U.S.P. Theophylline Reference Standard into a 200-ml volumetric flask. Dissolve in and dilute to volume with 0.1N sodium hydroxide solution. Pipet 3.0 ml into a second 200-ml volumetric flask and dilute to volume with 0.1N sodium hydroxide solution.

Accurately weigh the equivalent of approximately 190 mg of theophylline from the sample preparation powder into a 250-ml volumetric flask, add 100 ml of 0.1N sodium hydroxide solution and shake mechanically for 5 minutes. Dilute to volume with 0.1N sodium hydroxide solution and permit to stand for 10 minutes. Filter the supernatant solution through glass fiber filter paper (Whatman GF/C grade or equivalent), discarding the first 20 ml of filtrate. Pipet 3.0 ml of solution into a 250-ml volumetric flask and dilute to volume with 0.1N sodium hydroxide solution.

Determine the absorbances of the standard and sample solutions in a 1-cm cell, with a suitable spectrophotometer, at the point of maximum absorption at 274 mu, using 0.1N sodium hydroxide as the blank.

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Calculation:

$$\frac{Au}{As}\left(\frac{\text{mg Standard}}{200}\times\frac{3}{200}\right)\left(\frac{250}{\text{mg Sample}}\times\frac{250}{3}\right)\times\frac{\text{Average tablet wt (mg)}}{1}$$

= mg Theophylline per tablet

$$\frac{\text{mg Theophylline}}{130}\times 100 = \text{Percent required amount.}$$

Where: Au = absorbance of sample
As = absorbance of standard

4.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing one hundred (100) tablets of size 1 or five hundred (500) tablets of size 2 as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate containers (all sizes). Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. In addition, size 1 shall have a child-proof closure.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue of size 1 shall be packaged as specified in 5.2.7.1 of PPP-C-186. Twelve (12) units of issue of size 2 shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue of size 1 or forty-eight (48) units of issue of size 2 shall be packed as specified in 5.4.4 of PPP-C-186.

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5.3.2 Level B. One hundred forty-four (144) units of issue of size 1 shall be packed as specified in 5.4.3.1.1 of PPP-C-186. Forty-eight (48) units of issue of size 2 shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container (Size 1). The label of the immediate container for size 1 shall be a tri-part label consisting of the following:

(1) a tab portion shall remain attached to the immediate container. This portion shall bear:

(a) the National Drug Code (NDC)
(b) the lot or control number
(c) the expiration date

Note: Other label information may be included.

(2) A tear-off portion adjacent to the tab portion or at the extreme end of the label bearing the following:

(a) the National Drug Code (NDC)
(b) the lot or control number
(c) the expiration date

No other labeling shall appear thereon.
This portion shall be easily removed for attachment to a prescription.

(3) the principal portion connected to the tear-off and/or the tab portion shall bear the following:

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(a) the National Stock Number (which shall not be preceded by any prefix).
(b) all other information required by the Federal Food, Drug and Cosmetic Act.

5.4.1.2 Immediate container (Size 2). Each immediate container label for sizes 2 shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.3 Unit package. Each unit package label for sizes 1 and 2 shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

Notes: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied for size 2, the National Stock Number shall appear on the immediate container. If a unit package is not supplied for size 1, the National Stock Number shall appear on the principal portion connected to the tear-off and/or the tab portion of the label on each immediate container.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock No.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-149-0114	THEOPHYLLINE, EPHEDRINE SULFATE, AND HYDROXYZINE HYDROCHLORIDE TABLETS, 100s
6505-00-931-4329	THEOPHYLLINE, EPHEDRINE SULFATE, AND HYDROXYZINE HYDROCHLORIDE TABLETS, 500s

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1383
cs

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Philadelphia, PA. 19101

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DoD - 304



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HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
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PHILADELPHIA, PA 19101

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-T-37228 THEOPHYLLINE, EPHEDRINE SULFATE AND HYDROXYZINE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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